

MAT 320 HW11

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Due 12/10

Problem 1. Let φ be the step function

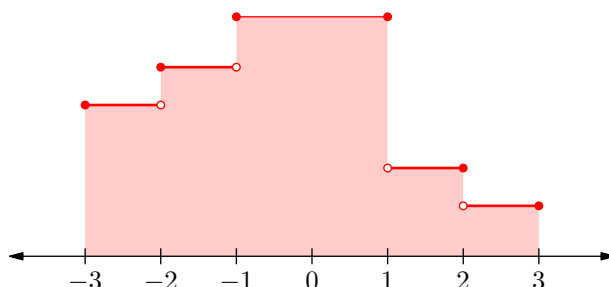
$$\varphi := 2 \cdot \varphi_{[-3,1]} + \frac{1}{2} \cdot \varphi_{[-2,2]} + \frac{2}{3} \cdot \varphi_{[-1,3]}.$$

Find the discontinuities of φ and sketch its graph. Compute its integral.

Solution. We can write φ as a piecewise constant function as follows:

$$\varphi = \begin{cases} 2 & -3 \leq x < -2 \\ 5/2 & -2 \leq x < -1 \\ 19/6 & -1 \leq x \leq 1 \\ 7/6 & 1 < x \leq 2 \\ 2/3 & 2 < x \leq 3 \\ 0 & \text{otherwise.} \end{cases}$$

Thus the graph of φ on the interval $[-3, 3]$ is



with discontinuities at $x \in \{-3, -2, -1, 1, 2, 3\}$. In particular the function is continuous at all other points because it is constant on the intervals $(-\infty, -3)$, $(-3, -2)$, $(-2, -1)$, $(-1, 1)$, $(1, 2)$, $(2, 3)$, $(3, \infty)$, so in particular it is constant in some neighborhood of any point outside the given discontinuities.

By definition, the integral of the step function φ is just

$$\begin{aligned} \int_{-\infty}^{\infty} \varphi &= 2 \int_{-\infty}^{\infty} \varphi_{[-3,1]} + \frac{1}{2} \int_{-\infty}^{\infty} \varphi_{[-2,2]} + \frac{2}{3} \int_{-\infty}^{\infty} \varphi_{[-1,3]} \\ &= 2(1 - (-3)) + \frac{1}{2}(2 - (-2)) + \frac{2}{3}(3 - (-1)) \\ &= 4 \left(2 + \frac{1}{2} + \frac{2}{3} \right) \\ &= 4 \cdot \frac{19}{6} = \boxed{\frac{38}{3}}. \end{aligned}$$

□

Problem 2. Show that the function $f(x) = x^2$ is integrable on $[0, 1]$ by approximating with step functions. That is, show that, for every $\varepsilon > 0$, there are step functions ψ and ϕ such that $\phi \leq f \leq \psi$ and

$$\int \psi - \int \phi < \varepsilon.$$

Solution. Let ϕ_n be the step function equaling $\left(\frac{k-1}{n}\right)^2$ on the interval $((k-1)/n, k/n]$ and let ψ_n be the step function equaling $\left(\frac{k}{n}\right)^2$ on the interval $((k-1)/n, k/n]$, where $k \in \{1, 2, \dots, n\}$, and let $\phi_n(x) = \psi_n(x) = 0$ everywhere else. We can see that, for $x \in [0, 1]$, either $x = 0$ or $(k-1)/n < x \leq k/n$ for some integer $1 \leq k \leq n$, in which case

$$\phi_n(x) = \left(\frac{k-1}{n}\right)^2 < x^2 \leq \left(\frac{k}{n}\right)^2 \leq \psi_n(x).$$

That is, $\phi_n(x) \leq f(x) \leq \psi_n(x)$ for all $x \in [0, 1]$. Moreover, we can see that

$$\psi_n(x) - \phi_n(x) = \left(\frac{k}{n}\right)^2 - \left(\frac{k-1}{n}\right)^2 = \frac{2k-1}{n^2}$$

when $x \in ((k-1)/n, k/n]$, so the integral of the difference between the functions is equal to

$$\sum_{k=1}^n \frac{1}{n} \cdot \frac{2k-1}{n^2} = \frac{1}{n^3} \cdot (2 \cdot (1 + 2 + \dots + n) - n) = \frac{n^2}{n^3} = \frac{1}{n}.$$

Thus, for each ε , pick n large enough so that $\frac{1}{n} < \varepsilon$, and then take $\psi := \psi_n$ and $\phi := \phi_n$ to get that $\phi \leq f \leq \psi$ and

$$\int \psi - \int \phi < \varepsilon,$$

which proves that x^2 is Riemann integrable. $\square$

Problem 3. Show that the function

$$f(x) := \begin{cases} 1 & x \in \mathbb{Q} \cap [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

is not Riemann integrable.

Solution. Consider an arbitrary step function ψ that bounds f from above. Then, any interval $I \subseteq [0, 1]$ of nonzero length contains some rational number. If $I_1, I_2, \dots, I_n$ are a partition of $[0, 1]$ such that ψ is constant on I_k for each $1 \leq k \leq n$, then $\psi \geq 1$ on each of the intervals with nonzero length (whose total length is thus still one) because, if $q_k \in \mathbb{Q} \cap I_k$, then $\psi(q_k) \geq f(q_k) = 1$. Thus,

$$\int \psi = \sum_{k=1}^n \ell(I_k) \cdot c_k = \sum_{\substack{1 \leq k \leq n \\ \ell(I_k) > 0}} \ell(I_k) \cdot c_k \geq \sum_{\substack{1 \leq k \leq n \\ \ell(I_k) > 0}} \ell(I_k) \cdot 1 = 1.$$

On the other hand, by the same logic, if ϕ is a step function bounding f from below, then we can find that $\phi(x) \leq 0$ for all but finitely many $x \in [0, 1]$, so

$$\int \phi \leq 0.$$

Thus the infimum of the integrals of step functions from above cannot be equal to the supremum of the integrals of step functions from below, so f is not Riemann integrable. $\square$

Problem 5. Show that the function

$$f(x) = \begin{cases} 1/q & x = p/q \in \mathbb{Q} \cap (0, 1) \text{ with } \gcd(p, q) = 1 \\ 0 & \text{otherwise} \end{cases}$$

is Riemann integrable and has integral zero. Find all points at which f is continuous, too.

Solution. Consider the function $f_n(x)$ that is $f(x)$ restricted to rational numbers with denominator at most n (where it is understood that the denominator refers to the denominator when $x = p/q$ with $\gcd(p, q) = 1$ and $q > 0$). That is,

$$f_n(x) = \begin{cases} 1/q & x = p/q \in \mathbb{Q} \cap (0, 1) \text{ with } \gcd(p, q) = 1 \text{ and } q \leq n \\ 0 & \text{otherwise} \end{cases}$$

Claim 5.1 — The f_n converge uniformly to f .

Proof. Indeed, observe that

$$|f(x) - f_n(x)| = \begin{cases} 1/q & x = p/q \in \mathbb{Q} \cap (0, 1) \text{ with } \gcd(p, q) = 1 \text{ and } q > n \\ 0 & \text{otherwise} \end{cases}$$

because the only points at which f and f_n disagree are rational numbers in $(0, 1)$ whose denominators are more than n . In particular we get

$$d(f, f_n) = \sup_{x \in \mathbb{Q} \cap (0, 1)} |f(x) - f_n(x)| = \frac{1}{n+1},$$

which goes to zero as n grows arbitrarily large, independently of x , so $f_n \rightarrow f$ uniformly. $\square$

Now we know that the uniform limit of Riemann integrable functions is also Riemann integrable, and that their integrals converge as expected. Since each f_n is zero except at finitely many points, each f_n has an integral of zero; thus f also has an integral of zero.

I now claim that the set of points at which f is continuous is all the points at which $f(x) = 0$, i.e. $\mathbb{R} \setminus (\mathbb{Q} \cap (0, 1))$. Clearly f is discontinuous on $\mathbb{Q} \cap (0, 1)$ because if $f(x) = 1/q$ then, picking $\varepsilon < \frac{1}{q} - \frac{1}{q+1}$, there is no $y \in \mathbb{R}$ for which $|f(x) - f(y)| < \varepsilon$.

We can argue that f is continuous at the irrational points of $(0, 1)$ with the following argument:

Claim 5.2 — If a sequence of rationals (q_n) converges to an irrational number r , then the denominators of the q_n grow arbitrarily large.

Proof. Suppose that the denominators are bounded above by N . There are only finitely many rational numbers with denominator at most N within the interval $(r-1, r+1)$, and all q_n must of course eventually stay in this interval. By the Pigeonhole principle, one of these rational numbers appears infinitely often among the (q_n) , and, supposing that the q_n converge, this rational number must be the limit of the q_n — but then the limit r is rational, contradiction. $\square$

Thus any sequence of rationals (q_n) converging to an irrational r has $\lim f(q_n) = \lim \frac{1}{q_n} = 0$. From here we can easily conclude that f is sequentially continuous on $(0, 1) \setminus \mathbb{Q}$.

For $(\infty, 0] \cup [1, \infty)$, we only need to verify continuity at zero and one. It is easy to show that the claim above holds when r is an integer as long as we restrict $q_n \notin \mathbb{Z}$, since the distance between a non-integral rational p/q and an integer is bounded below by $1/q$, so we can conclude sequential continuity at zero from the right and at one from the left; f being constantly zero on the rest of $\mathbb{R}$ finishes. $\square$

Problem 7. Show that if f is Riemann integrable on $[a, b]$ then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

for all $c \in (a, b)$.

Solution. Observe that

$$\int_a^c f = \int_a^b f \cdot \varphi_{[a,c]}, \quad \int_c^b f = \int_a^b f \cdot \varphi_{[c,b]},$$

and that

$$f \cdot \varphi_{[a,c]} + f \cdot \varphi_{[c,b]} = f \cdot \varphi_{[a,b]},$$

so we can use the fact that the integral is a linear operator to say

$$\int_a^b f = \int_a^b f \cdot \varphi_{[a,c]} + f \cdot \varphi_{[c,b]} = \int_a^b f \cdot \varphi_{[a,c]} + \int_a^b f \cdot \varphi_{[c,b]} = \int_a^c f + \int_c^b f. \quad \square$$

Problem 8. Show that if f is a nonnegative continuous function such that

$$\int_a^b f(x) dx = 0$$

then $f(x) = 0$ for all $x \in [a, b]$.

Proof. Suppose $f(r) > 0$ for some $r \in [a, b]$. Then, picking $\varepsilon < f(r)$, we can find a δ for which $f(x) > f(r) - \varepsilon > 0$ when $r - \delta < x < r + \delta$ because of the fact that f is continuous. Since $r \in [a, b]$, the intersection of $(r - \delta, r + \delta)$ with $[a, b]$ has nonzero length, say d , so we can compute

$$\int_a^b f \cdot \varphi_{(r-\delta, r+\delta)} \geq d(f(r) - \varepsilon) > 0.$$

Then, we know that f is nonnegative on $[a, b] \setminus (r - \delta, r + \delta)$, so we can conclude that $f \geq f \cdot \varphi_{(r-\delta, r+\delta)}$ everywhere since the latter function is zero outside $(r - \delta, r + \delta)$, which gives us the inequality

$$\int_a^b f \geq \int_a^b f \cdot \varphi_{(r-\delta, r+\delta)} > 0$$

as long as there exists a point $r \in [a, b]$ at which f is positive.

Therefore, if the integral of f is equal to zero, such an r cannot exist, i.e. f is the zero function. $\square$

Problem 11. Show that the equation

$$x^2 = x \sin x + \cos x$$

has exactly two solutions.

Solution. Define $f(x) = x^2 - x \sin x - \cos x$. We can see that f is an even function, i.e.

$$f(-x) = (-x)^2 - (-x) \sin(-x) - \cos(-x) = x^2 - x \sin x - \cos x = f(x),$$

so $f(r) = 0$ iff $f(-r) = 0$. Note that $f(0) = -1 < 0$.

Claim 11.1 — f is strictly increasing on $(0, \infty)$.

Proof. We can compute

$$f'(x) = 2x - x \cos x - \sin x + \sin x = 2x - x \cos x.$$

Then, since $x > 0$, we can bound

$$1 \cdot x < (2 - \cos x) \cdot x < 3 \cdot x,$$

so in particular $f'(x) > 0$ when x is positive. $\square$

We can see that

$$f(0) < 0 < f(\pi) = \pi^2 + 1,$$

so by the intermediate value theorem f is zero at a real number $c \in (0, \pi)$, and f being increasing on $(0, \infty)$ means that $f(x) < 0$ for $0 < x < c$ and $f(x) > 0$ for $x > c$, so in particular f is zero at exactly one point in $(0, \infty)$. As stated earlier, f being even means that there is also exactly one negative real at which $f(x) = 0$, namely $x = -c$, so $f(x) = 0$ has only two solutions $x = \pm c$. $\square$

Problem 12. Find all differentiable functions $f: (1, \infty) \rightarrow \mathbb{R}$ such that

$$f(xy) = f(x) + f(y)$$

for all $x, y > 1$.

Solution 1. In fact we don't need differentiability, only continuity.

The only such functions f are of the form $c \cdot \log x$ for any real c . Indeed, we can see that such f work because

$$f(xy) = c \log(xy) = c(\log x + \log y) = c \log x + c \log y = f(x) + f(y).$$

Now we show that all f must be of this form. To start with, note that if $x > 1$, then $x^\alpha > 1$ for all $\alpha > 0$, so as long as $x > 1$, we can pass x^α as an input of f . We can see first that, for any $x > 1$ and positive integer n ,

$$f(x^n) = f(x^{n-1}) + f(x) = f(x^{n-2}) + f(x) + f(x) = \cdots = nf(x).$$

Next, we can get that for any two positive integers m, n ,

$$nf(x^{m/n}) = f((x^{m/n})^n) = f(x^m) = mf(x) \implies f(x^{m/n}) = \frac{m}{n}f(x).$$

Thus if we are given $f(x)$, we can compute $f(x^q) = qf(x)$ for any positive rational q .

Now fix $x > 1$ and let $\alpha > 0$ be arbitrary. Pick a sequence of positive rationals (q_n) converging to α . Then, since f is differentiable, it is continuous, so, using the continuity of the map $t \mapsto x^t$,

$$f(x^\alpha) = \lim_{n \rightarrow \infty} f(x^{q_n}) = \lim_{n \rightarrow \infty} q_n f(x) = \alpha f(x).$$

To finish, any element of $(1, \infty)$ can be written as some e^y with $y = \log x > 0$, so we get that for any $x > 1$,

$$f(x) = f(e^y) = y \cdot f(e) = \log x \cdot f(e),$$

where $f(e)$ is some constant. That is, any function satisfying the given constraints must be of the form $c \cdot \log x$ for some real c , completing the proof. $\square$

Solution 2. Differentiate both sides with respect to x to get

$$y f'(xy) = f'(x)$$

and with respect to y to get

$$x f'(xy) = f'(y).$$

Then, dividing, we conclude that

$$\frac{f'(x)}{f'(y)} = \frac{y}{x}.$$

Now, fixing y , we see that $y f'(y)$ is just a constant, so

$$f'(x) = \frac{c}{x}.$$

Taking for granted the fact that the integral of $\frac{1}{x}$ is $\log x$, this implies that

$$f(t) = c \log t - c + f(e) = c \log t + d$$

for some constants c and $d := f(e) - c$ (integrate from e to t). We can check that $d = 0$, as

$$c \log(xy) + d = f(xy) = f(x) + f(y) = c \log x + c \log y + 2d = c \log(xy) + 2d.$$

Thus f must be of the form $c \log x$, and indeed c can be any real number, as checked in Solution 1, so the full solution set is $f(x) = c \log x$ for any $c \in \mathbb{R}$. $\square$